

Topic 106: Practice Assignment- Data Modelling in Power BI

Using your Adventure Works report file, complete the following:

1) Navigate to the **MODEL** view, and perform the following actions

- Right-click to delete each relationship between **Sales Data**, **Customer_Lookup** and **Calendar_Lookup** (*including both date fields*)
- Use the **Manage Relationships** tool to delete *all* remaining relationships between all tables

2) Recreate all table relationships (*using any method you prefer*), and confirm the following:

- Cardinality is **1-to-Many** for all relationships
- Filters are all **One-Way** (*no two-way filters*)
- Filter direction correctly flows "**downstream**" to data tables
- Data tables are **not connected** directly to one another
- Both data tables are connected to **all valid lookup tables**
- Product-related tables follow a **snowflake schema**

3) Return to the **REPORT** view, and complete the following:

- Edit (*or insert*) the matrix visual to show **ReturnQuantity** (*values*) by **CategoryName** (*rows*) from the **Product_Category_Lookup** table
 - *Which category saw the highest volume of returns? How many?*
- Replace **CategoryName** with **Year** from the **Calendar_Lookup** table
 - *How many returns do you see in 2015 vs. 2016?*
- Replace **Year** with **FullName** from the **Customer_Lookup** table
 - *What do you see, and why?*
- Update the matrix to show both **OrderQuantity** and **ReturnQuantity** (*values*) by **ProductKey** (*rows*) from the **Product_Lookup** table
 - *What was the total OrderQuantity for Product #338?*

4) Unhide the **ProductKey** field from the **Returns** tables (*using either the **DATA** or **RELATIONSHIPS** view*):

- In the matrix, replace **ProductKey** from **Product_Lookup** with **ProductKey** from the **Returns Data** table
 - *Why do we see the same repeating values for OrderQuantity?*
- Edit the relationship between **Returns Data** and **Product_Lookup** to change the cross filter direction from *Single* to *Both*
 - *Why does the visual now show OrderQuantity values by product, even though we are using **ProductKey** from **Returns Data**?*
 - *How many orders do we see now for Product #338? What's going on here?*

5) Complete the following:

- Change the cross filter direction between **Returns Data** and **Product_Lookup** back to *single* (*One-Way*)
- Hide the **ProductKey** field in the **Returns Data** table from report view (*and any other foreign keys, if necessary*)
- Update the matrix to show **ProductKey** from the **Product_Lookup**, rather than **Returns Data**

6) Save the Power BI File